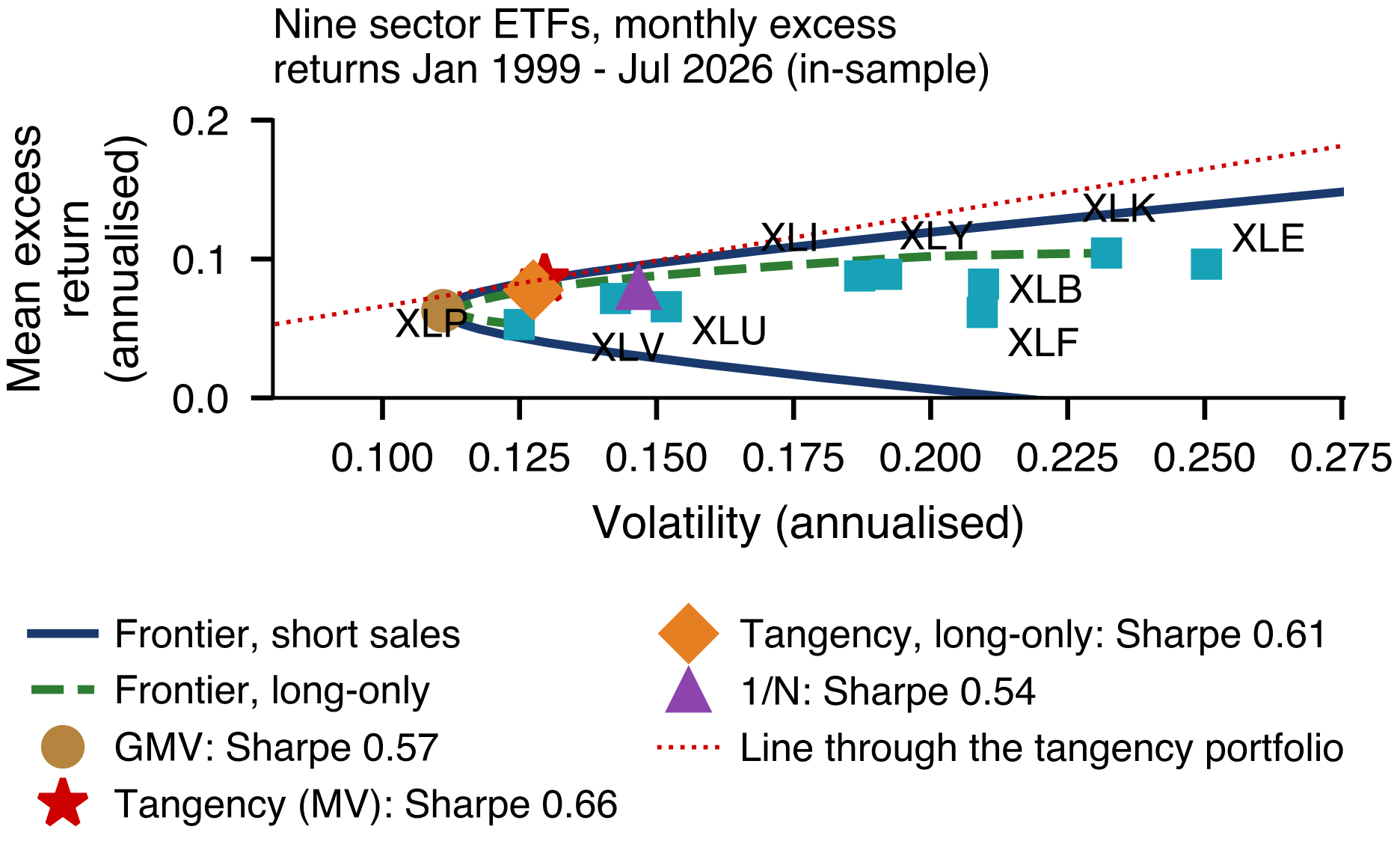
Nine sector ETFs, monthly excess returns Jan 1999 - Jul 2026 (in-sample)
Mean excess return (annualised)
0.2
0.1
0.0
0.100
0.125
0.150
0.175
0.200
0.225
0.250
0.275
Volatility (annualised)
XLP
XLV
XLU
XLI
XLY
XLK
XLB
XLF
XLE
Frontier, short sales
Frontier, long-only
GMV: Sharpe 0.57
Tangency (MV): Sharpe 0.66
Tangency, long-only: Sharpe 0.61
1/N: Sharpe 0.54
Line through the tangency portfolio